Pharmacist Module: Fix, Rebuild and Demo-Ready

Task brief for Claude Code (VS Code) - Vernex Clinic OS - prepared 13 Oct 2026 from a read-only analysis of the current codebase

How to use this document. Save this PDF in the repo as `docs/tasks/PHARMACIST_MODULE_TASK.pdf`, open Claude Code in the project root and paste the starter prompt from Section 0. Claude Code should work through Tasks 1 to 10 **in order**, finishing each task's acceptance checks before starting the next. Every finding below names the real file and the real cause, so no re-discovery is needed - but Claude must still read each file before editing it.

0. Starter prompt (paste this into Claude Code)

```
Read docs/tasks/PHARMACIST_MODULE_TASK.pdf completely before changing anything.
It is the full specification for fixing and rebuilding the pharmacist module of
this repo (React + Vite frontend in frontend/, Supabase in supabase/).

Work through Tasks 1-10 strictly in order. For each task:
  1. Read every file the task names before editing it.
  2. Implement exactly what the task says, matching the surrounding code style
     (services layer, ui components, Tailwind tokens, toast via sonner).
  3. Run `npm run build` and `npm run lint` inside frontend/ and fix every error.
  4. Tick off that task's acceptance checks and tell me what you verified.
Never edit an existing migration file - add new, idempotent migrations.
Never use or ask for the Supabase service-role key; never commit frontend/.env.
Do not push to the remote database yourself: when a task adds SQL, stop and
tell me exactly which file to apply (supabase db push, or paste in SQL editor).
If something in the spec conflicts with the code you find, stop and ask me.
```

Ground rules for every task

- **Live data only.** Screens read and write through `services` (`frontend/src/services/serviceProvider.ts`). No mock arrays in pages.
- **Database is the authority.** Any new write path that touches several tables (stock purchase, Excel import, auto pharmacy order) must be a single `security definer` RPC so it is atomic and permission-checked server-side.
- **Migrations:** new files only, timestamp prefix after `20260919120000`, e.g. `20261013120000_pharmacist_access_fix.sql`. Use `create or replace`, `if not exists`, `drop policy if exists` and `on conflict` so they can be re-run.
- **Types:** after adding RPCs/columns, update `frontend/src/shared/types/database.types.ts` (regenerate with the Supabase CLI if available, otherwise hand-add the exact `Functions`/`Tables` entries).
- **Money and dates:** show rupees with the existing `rupee()`/`formatCurrency` helpers. Compute "today" in the clinic's local time (IST), not with `new Date().toISOString().slice(0,10)`, which is UTC and is wrong between 00:00 and 05:30 IST. Add one shared helper (e.g. `localDateISO()` in `frontend/src/utils/formatDate.ts`) and use it in all pharmacy pages.
- **Mobile:** every new page must work at 375 px width (tables become cards or scroll horizontally inside their card).

1. Analysis: why the pharmacist experience is broken

These are confirmed from the code. Severity: **P0** = blocks login/usage, **P1** = feature broken, **P2** = quality.

#	Sev	Finding	Where
F1	P0	The pharmacist role is never granted **pharmacy.manage**. Phase 1 grants pharmacist only view/create/edit/assign/cancel/bill/dispense/configure. But the Medicine Stock and Purchase Entry routes and nav items require `manage`, so they are hidden and opening them is denied.	supabase/migrations/20260730191500_phase_1_backend_foundation.sql (role_permissions insert near line 258); frontend/src/routes/AppRoutes.tsx (/pharmacy/stock, /pharmacy/purchase-entry); frontend/src/components/layout/navigation.ts
F2	P0	**Per-staff rows override role rows.** `buildAuthContext()` uses staff_permissions/staff_modules if any exist, and `has_permission()` does `coalesce(staff, role)`. The demo seed copies role permissions into staff_permissions at seed time, so fixing role_permissions alone will NOT fix an already-seeded pharmacist.	frontend/src/services/supabase/supabaseAuth.service.ts (buildAuthContext); phase_1 has_permission(); supabase/seed/demo_seed.sql section 2
F3	P0	**Every access denial redirects to /login.** ModuleGuard sends a denied (but signed-in) user to /login, so the pharmacist appears to be "logged out" / "login not working" when they hit a page their permissions do not cover. There is no access-denied screen.	frontend/src/access-control/ModuleGuard.tsx (Navigate to /login); frontend/src/components/layout/AppLayout.tsx
F4	P0	**Login fails opaquely when the profile is missing.** If `pharmacist@vernex.test` was created in Auth after the seed ran (seed skips users that do not exist yet), there is no staff_profiles row. `loadCurrentStaffProfile()` uses `.single()` and throws "JSON object requested, multiple (or no) rows returned". The Supabase session is also left signed-in after the failure.	supabaseAuth.service.ts (loadCurrentStaffProfile, signInWithPassword); demo_seed.sql section 2
F5	P1	**Prescription Queue and Counter Billing are missing from the pharmacist sidebar.** The routes exist (/pharmacy/prescriptions, /pharmacy/billing) but the nav never links them, so doctor prescriptions look like they never arrive.	navigation.ts (navForRole.pharmacist)
F6	P1	**Pharmacy orders are created lazily.** A routed prescription only becomes a pharmacy_orders row when the pharmacist opens the Prescription Queue page (its load() calls create_pharmacy_order_from_prescription for each). The dashboard never does this, so new prescriptions do not show on the dashboard.	frontend/src/modules/pharmacy/pages/PrescriptionQueue.tsx (load); PharmacyDashboard.tsx
F7	P1	**Doctor must click a separate "Send to Pharmacy" button.** Save and Complete Consultation never route. Toasts still say "placeholder". Many prescriptions therefore never reach pharmacy.	frontend/src/modules/doctor/components/PrescriptionBuilder.tsx (routeToPharmacy, finalizePrescription)
F8	P1	**Medicine matching is exact-name only.** Order items match `lower(medicines.name) = lower(prescription_items.medicine_name)`. Doctor's free text ("Paracetamol 650" or "Dolo 650") becomes 'unavailable' even when stocked. Doctor items never carry medicine_id (autocomplete returns names only).	phase_9 create_pharmacy_order_from_prescription; frontend/src/modules/doctor/components/MedicineSearchInput.tsx
F9	P1	**Purchase entry overwrites stock.** createPurchaseEntry upserts on (medicine_id, batch_no) with `quantity_available: item.quantity`, so receiving more of an existing batch REPLACES the quantity instead of adding. It is also many client-side writes with no transaction.	frontend/src/services/supabase/supabaseCatalog.service.ts (createPurchaseEntry, ~line 171-250)
F10	P2	No Excel/CSV stock upload exists. No xlsx library is installed.	frontend/package.json
F11	P2	Dashboard and Medicine Stock call getMedicineById once per medicine (N+1 requests).	PharmacyDashboard.tsx, MedicineStock.tsx
F12	P2	Mojibake text on the login page ("Ã‚Â·") and in WhatsAppBookingPanel ("Ã¢â‚¬Å“").	frontend/src/pages/auth/Login.tsx line 30; frontend/src/modules/reception/components/WhatsAppBookingPanel.tsx

Most likely real-world cause of "pharmacist login not working": F4 (no staff profile because the Auth user was created after the seed) or F1+F2+F3 (signed in, but the first permission check fails and ModuleGuard bounces back to /login). Task 1 and Task 2 fix all of these together; Task 1 also gives SQL to confirm which one applied.

Task 1 - Database: fix pharmacist access (P0)

Create `supabase/migrations/20261013120000_pharmacist_access_fix.sql`. It must:

1. Grant role **pharmacist** these permissions in role_permissions (allowed = true): `pharmacy.view`, `pharmacy.create`, `pharmacy.edit`, `pharmacy.manage`, `pharmacy.dispense`, `pharmacy.bill`, `pharmacy.export`, `prescriptions.view`, `billing.view`, `billing.create`, `billing.bill`, `dashboard.view`. Ensure role_modules has dashboard, pharmacy, prescriptions, billing enabled.
2. Bring **every existing pharmacist staff member** up to date (fixes F2): upsert the same permission keys into staff_permissions and the same module keys into staff_modules for every `staff_profiles` row with role_key = 'pharmacist'. Use `on conflict ... do update set allowed = true`. Do not remove any existing grants.
3. Do not touch other roles and do not enable clinic_modules for real clinics automatically (the demo seed does that for the demo clinic).

Diagnostic SQL

Claude Code must also save this as `supabase/seed/diagnose_pharmacist.sql` (read-only queries) and tell the user to run it in the Supabase SQL editor before and after the migration:

```
-- 1. Does the Auth user exist and is it confirmed?
select id, email, email_confirmed_at, last_sign_in_at
from auth.users where lower(email) = 'pharmacist@vernex.test';

-- 2. Is there an active staff profile linked to it?
select sp.id, sp.user_id, sp.role_key, sp.status, sp.clinic_id, c.slug, c.status as clinic_status
from public.staff_profiles sp left join public.clinics c on c.id = sp.clinic_id
where lower(sp.email) = 'pharmacist@vernex.test';

-- 3. Effective modules (staff rows override role rows) and clinic switch
select sm.module_key, sm.enabled, cm.enabled as clinic_enabled
from public.staff_profiles sp
join public.staff_modules sm on sm.staff_id = sp.id
left join public.clinic_modules cm on cm.clinic_id = sp.clinic_id and cm.module_key = sm.module_key
where lower(sp.email) = 'pharmacist@vernex.test' order by 1;

-- 4. Effective pharmacy / prescription / billing permissions
select sper.permission_key, sper.allowed
from public.staff_profiles sp join public.staff_permissions sper on sper.staff_id = sp.id
where lower(sp.email) = 'pharmacist@vernex.test'
  and split_part(sper.permission_key, '.', 1) in ('pharmacy','prescriptions','billing','dashboard')
order by 1;
```

Interpretation to write as comments in the file: query 1 empty = create the Auth user (Auto Confirm) and re-run the seed; query 2 empty = re-run the seed; query 3/4 missing pharmacy.manage = apply this migration.

Acceptance checks

- Migration runs twice in a row without error.
- Query 4 shows pharmacy.manage = true for the demo pharmacist after applying it.
- No permission changed for owner, admin, doctor, receptionist (verify with a count query before/after).

Task 2 - Frontend: make login and route guarding robust (P0)

Files: `supabaseAuth.service.ts`, `pages/auth/Login.tsx`, `access-control/ModuleGuard.tsx`, `routes/AppRoutes.tsx`, `components/layout/AppLayout.tsx`.

1. **loadCurrentStaffProfile:** use `.maybeSingle()`. If no row, throw a clear Error: "Your account is signed in but not linked to a clinic staff profile. Ask your clinic admin to add you." If status is not 'active', throw "Your staff account is inactive." If the role is not super_admin and the clinic is missing or not active, throw "Your clinic is not active."
2. **signInWithPassword:** if building the auth context fails after Supabase sign-in succeeded, call `supabase.auth.signOut()` and clear the local cache before re-throwing, so the user is never left half signed in. Map Supabase errors to friendly text: "Invalid login credentials" -> "Email/staff ID or password is incorrect."; "Email not confirmed" -> "This account has not been confirmed yet."; staff-ID lookup returning null -> "No active staff member with this staff ID."
3. **Login.tsx:** replace the mojibake with " · " separators (and fix WhatsAppBookingPanel the same way, using proper quotes and an arrow). If a valid session already exists when /login loads, redirect straight to `homeForRole[role]`.
4. **Access denied screen:** create `frontend/src/pages/shared/AccessDenied.tsx` (uses PageHeader/EmptyState, shows the missing module/permission, a button "Go to my dashboard" -> homeForRole[role], and "Sign out"). In ModuleGuard: no session -> /login; session but denied -> render AccessDenied (do NOT navigate to /login). The guard must never cause a redirect loop.
5. Keep the existing 60-second auth-context cache, but invalidate it after login and on `refreshAuthContext()`.

Acceptance checks

- pharmacist@vernex.test and staff ID VNX-PHAR-001 both sign in and land on /pharmacy/dashboard.
- A pharmacist opening /owner/dashboard sees Access Denied with a working "Go to my dashboard" button, and stays signed in.
- An Auth user with no staff profile sees the friendly message and is signed out (check: refreshing /login does not auto-enter).
- Owner, receptionist and doctor logins still work exactly as before.

Task 3 - Pharmacist navigation and routes

Replace `navForRole.pharmacist` in `navigation.ts` with this order (use lucide icons already imported or add suitable ones):

Label	Path	Module / permission
Dashboard	/pharmacy/dashboard	pharmacy / view
Prescription Queue (with pending-count badge)	/pharmacy/prescriptions	prescriptions / view
Counter Billing	/pharmacy/billing	billing / bill
Medicine Stock	/pharmacy/stock	pharmacy / view
Add Stock (new)	/pharmacy/add-stock	pharmacy / manage
Low Stock	/pharmacy/low-stock	pharmacy / view
Expiry Alerts	/pharmacy/expiry-alerts	pharmacy / view

- In AppRoutes: add `/pharmacy/add-stock` (roles pharmacist, owner, admin; module pharmacy; permission manage). Change `/pharmacy/stock` to permission **view** (editing actions on that page are hidden unless the user has manage). Make `/pharmacy/purchase-entry` redirect to `/pharmacy/add-stock?tab=history`.
- Restrict /pharmacy/prescriptions and /pharmacy/billing to roles pharmacist, owner, admin (they are currently open to any role with the module).
- Pending badge: count of pharmacy_orders with status pending or partially_dispensed; refresh it with the realtime/poll mechanism from Task 7. Sidebar must support an optional `badge` on NavItem.

- Owner/admin nav: add a "Pharmacy" item -> /pharmacy/dashboard so owners can view pharmacy too (route roles must include owner/admin).

Acceptance: every link in the pharmacist sidebar opens a working page, none bounce to Access Denied.

Task 4 - Database: atomic stock intake RPC (fixes F9)

New migration `20261013121000_pharmacy_stock_intake.sql`. Add RPC `public.create_stock_purchase(input jsonb) returns jsonb`, security definer, search_path = public.

```
input = {
  "supplier_name": "MedSupply Distributors",   -- required
  "supplier_phone": "+918040001111",           -- optional
  "invoice_number": "PUR-2004",                -- required
  "purchase_date": "2026-10-13",               -- required
  "notes": "optional", "source": "manual" | "excel",
  "items": [ {
     "medicine_name": "Paracetamol 650mg",     -- required
     "generic_name": "Paracetamol", "category": "general", "manufacturer": "Cipla",
     "strength": "650mg", "unit": "tablet", "hsn_code": "3004", "gst_rate": 12,
     "reorder_level": 40,
     "batch_no": "PCM-A22",                    -- required
     "expiry_date": "2027-08-31",              -- required
     "quantity": 200,                          -- required, > 0
     "purchase_price": 1.80, "selling_price": 2.50, "mrp": 2.80
  } ]
}
returns { "purchase_id", "items": n, "medicines_created": n,
          "batches_created": n, "batches_topped_up": n, "total_amount": x }
```

- Permission: is_super_admin() OR (has_module('pharmacy') AND (has_permission('pharmacy.manage') OR has_permission('pharmacy.create'))). Clinic/branch come from the caller's staff profile, never from input.
- Validate everything first and raise one exception listing row numbers if any item is invalid: missing name/batch/expiry, quantity <= 0, negative prices, expiry date already past, selling_price > mrp when mrp > 0, the same (medicine, batch_no) twice in one input. Max 1000 items.
- Reject a duplicate invoice: same clinic + lower(supplier_name) + lower(invoice_number) already in stock_purchases -> exception "Invoice PUR-2004 from is already recorded".
- Medicine lookup: same clinic, active, `lower(btrim(name))` equal. If not found, insert a medicine with all provided master fields. If found, fill only empty master fields (do not overwrite pharmacist edits).
- Batch: insert, or on conflict (medicine_id, batch_no) **add** the quantity (`quantity_available = batch.quantity_available + excluded.quantity_available`), update prices/expiry/supplier/purchase_id, and set status 'active'.
- Insert stock_purchases (total_amount = sum(qty * purchase_price), status 'saved', metadata.source), one stock_movements row per item (movement_type 'add_stock', reference_type 'stock_purchase', reference_id purchase id, created_by auth.uid()), call refresh_low_stock_alerts once, write an audit_logs row.
- Grant execute to authenticated only. Rewrite `services.catalog.createPurchaseEntry` to call this RPC (keep its input type; map camelCase -> snake_case) and return the summary.

Acceptance: receiving 50 more of an existing batch with 12 in stock leaves 62; a failing row leaves the database unchanged; stock_movements has one row per item.

Task 5 - New "Add Stock" page: manual + Excel upload

Create `frontend/src/modules/pharmacy/pages/AddStock.tsx` with three tabs (use the existing `Tabs` component; remember the selected tab in the `?tab=` query param):

Tab A - Manual entry

- Header fields: supplier name (with suggestions from previous suppliers), supplier phone, invoice number, purchase date (default today, IST).

- Editable item grid reusing/refactoring `PurchaseEntryForm`: medicine (autocomplete from clinic medicines; picking one pre-fills generic, category, strength, unit, GST, last prices), batch no, expiry (month picker allowed, stored as last day of month), quantity, purchase price, selling price, MRP, reorder level. "New medicine" badge when the name is not in the catalogue, with extra fields (generic, manufacturer, strength, unit, HSN, GST).
- Live footer: item count, total units, invoice total (Rs.), and GST estimate. Add row / duplicate row / remove row. Keyboard friendly: Enter in the last cell adds a row.
- Inline validation identical to the RPC rules; Save disabled while invalid. On success show the RPC summary toast and reset.

Tab B - Excel / CSV upload

- Library: SheetJS, installed from the official CDN tarball (the npm-registry "xlsx" 0.18.5 has known vulnerabilities): `npm i https://cdn.sheetjs.com/xlsx-0.20.3/xlsx-0.20.3.tgz`. Load it with a dynamic `import("xlsx")` only on this tab so the main bundle does not grow.
- Accept .xlsx, .xls, .csv; max 5 MB; max 1000 data rows; read the first sheet (or a sheet named "Stock" if present). Drag-and-drop zone plus file picker.
- **Download template** button generates `vernex_stock_template.xlsx` in the browser: sheet "Stock" with the header row and 3 example rows, sheet "Instructions" explaining each column, accepted date formats and the rules.
- Supplier name, invoice number and purchase date are entered in form fields above the upload (one file = one supplier invoice).

Column (header)	Req.	Accepted header aliases (case/space-insensitive)	Rule
Medicine Name	Yes	medicine, item, item name, product, drug	text, trimmed
Batch No	Yes	batch, batch number, batch no., lot	text
Expiry Date	Yes	expiry, exp, exp date, expiry dt	Excel date serial, DD-MM-YYYY, DD/MM/YYYY, YYYY-MM-DD, MM/YY, MMM-YY; month-only -> last day of month; must be in future
Quantity	Yes	qty, units, quantity received	integer > 0
Purchase Price	Yes	rate, cost, purchase rate, ptr	number >= 0
Selling Price	Yes	sale price, selling rate, price	number >= 0
MRP	No	m.r.p	number; selling must not exceed it
Generic Name, Category, Manufacturer, Strength, Unit, HSN Code, GST %, Reorder Level	No	generic, company/mfr, form, hsn, gst, tax %, min stock	used only when the medicine is new; category must be one of the app's MedicineCategory values (else 'general' with a warning)

- **Preview step** (before anything is saved): table of parsed rows with a status chip per row - Valid, Warning (e.g. new medicine will be created, expiry within 90 days, category defaulted), Error (with the exact reason). Summary bar: "48 valid - 2 warnings - 3 errors". Rows are editable inline and removable. Unknown columns are listed and ignored. Duplicate (medicine, batch) rows in the file are flagged.
- Import button: "Import 50 rows", disabled while errors remain unless the user ticks "Skip rows with errors". Sends one `create_stock_purchase` call with source 'excel'. Show a result card with the summary and a "Download error report" (CSV of skipped rows with reasons).
- Put parsing/validation in a pure module `frontend/src/modules/pharmacy/stockImport.ts` (header mapping, date parsing, row validation) so it is easy to test; the page only renders.

Tab C - History

- List of stock_purchases (newest first) with date, supplier, invoice, items, total, source (Manual / Excel). Click opens a drawer with the batches it created (join medicine_stock_batches.purchase_id). Filter by date range and supplier. Export CSV.

Also add `docs/sample-data/pharmacy_stock_upload_sample.csv` with 25 realistic rows (mix of existing demo medicines and 5 new ones) so the upload can be demonstrated immediately.
Acceptance: the sample CSV and a template-generated .xlsx both import correctly; a file with a past expiry, a text quantity and a missing batch shows 3 precise errors and imports nothing until fixed or skipped.

Task 6 - Medicine Stock page upgrade

- One row per **medicine** (not per batch): name + generic + strength, category, total available units (active, non-expired batches), nearest expiry, batch count, selling price, stock status chip (In stock / Low / Out / Expired). Expand a row to see its batches (batch, expiry, qty, purchase/selling/MRP, supplier, status).
- Actions (only with pharmacy.manage): Edit medicine (master fields + reorder level), Adjust stock on a batch (existing StockAdjustmentModal -> adjust_medicine_stock), Mark batch blocked, View movement history (drawer listing stock_movements for that medicine with type, qty, reason, who, when). Primary page button "Add stock" -> /pharmacy/add-stock.
- Filters: search (name/generic/batch/supplier), category, stock status, expiring within (30/60/90 days). Sort by name, stock, expiry. Export CSV of the current view.
- Performance (F11): add one service method `getMedicineCatalog()` that loads full medicine rows in a single query (`select('*')`) and use it on this page and the dashboard instead of getMedicineById per row.

Task 7 - Make doctor prescriptions reliably reach the pharmacist

Database (new migration `20261013122000_prescription_pharmacy_flow.sql`)

1. Split create_pharmacy_order_from_prescription into an internal function `public._ensure_pharmacy_order(p_prescription_id uuid)` (no permission check, idempotent, revoke execute from public/anon/authenticated) and keep the public RPC as a permission-checked wrapper with the same signature.
2. Better medicine matching inside it, in this priority: prescription_items.medicine_id -> exact lower(btrim(name)) -> normalised name (lowercase, remove spaces, dots and hyphens) -> name that starts with the prescribed text -> generic_name match. Only pick active medicines in the same clinic; prefer the one with available stock. Still mark 'unavailable' when nothing matches or total stock is 0, with a note saying which.
3. Trigger `after insert or update of send_to_pharmacy` on prescriptions: when send_to_pharmacy becomes true, call _ensure_pharmacy_order(new.id). Note: create_prescription_with_items inserts items after the header, so the trigger must handle the case where items arrive later - simplest correct approach: also call _ensure_pharmacy_order at the end of route_prescription_to_pharmacy and at the end of create_prescription_with_items when send_to_pharmacy is true, and have _ensure_pharmacy_order add any missing items to an existing pending order.
4. Backfill: for every prescription with send_to_pharmacy = true and no pharmacy order, create one (inside the migration).
5. Realtime: add pharmacy_orders to the supabase_realtime publication if not already in it (guard with a check on pg_publication_tables).

Doctor side - PrescriptionBuilder.tsx

- Add a "Send to pharmacy" checkbox next to the action bar, **checked by default**. "Complete Consultation" = save (if needed) + finalize + route when checked. Keep the separate button for routing without finalizing.
- Remove every "placeholder" wording from pharmacy toasts: "Prescription sent to pharmacy." Show a status chip after routing: Sent to pharmacy -> Received -> Dispensed (read from prescriptions.pharmacy_status).
- Doctor MedicineSearchInput: return the medicine id as well as the name when a catalogue medicine is chosen, and send it as prescription_items.medicine_id. Show available stock next to each suggestion ("in stock 180" / "out of stock") so the doctor knows before prescribing.

Pharmacist side - Prescription Queue page

- Remove the lazy createOrderFromPrescription loop from load() (the DB now creates orders). Keep a manual "Sync" fallback that calls it for routed prescriptions without orders.
- Tabs: Pending | Partially dispensed | Dispensed today | All. Each card/row: token (short order number), patient name + age/gender + phone, doctor, diagnosis, time since routed, item count, unavailable-item warning.
- Detail drawer per order: for each item show requested qty, dosage/frequency/duration, available total stock, a batch selector defaulting to **FEFO** (earliest non-expired expiry with enough stock), editable dispense quantity, and "Mark unavailable". Items with no stock are clearly red. Buttons: Dispense selected, Dispense and create bill (calls dispense then billing.createInvoiceFromPharmacyOrder, then opens the bill), Print dispensing slip.
- Show patient allergies prominently (patients.allergies) in the drawer header.
- Live updates: subscribe with `supabase.channel(...).on('postgres_changes', {event:'*', schema:'public', table:'pharmacy_orders', filter:'clinic_id=eq.<id>'})`; also poll every 30 s as a fallback; show a toast "New prescription from Dr. X" for inserts. Put this in a hook `usePharmacyQueueLive()` used by the queue page, the dashboard and the sidebar badge. Unsubscribe on unmount.

Acceptance (end-to-end, must be demonstrated)

1. Doctor (doctor@vernex.test) writes a prescription for a demo patient with Paracetamol 650mg and Amoxicillin 500mg and clicks Complete Consultation.
2. Within 30 s, without refreshing, the pharmacist sees it on the dashboard and in the queue with a toast, and the sidebar badge increments.
3. Pharmacist dispenses with FEFO batches; stock of those batches drops by the dispensed quantity; stock_movements has 'dispense' rows.
4. Doctor's prescription status shows Dispensed. A prescribed name not in the catalogue shows as Unavailable, not an error.

Task 8 - Pharmacist dashboard polish

- KPI tiles (StatsCard): Pending prescriptions, Dispensed today, Today's sales (Rs., pharmacy invoices), Low stock items, Expiring in 30 days, Expired stock value (Rs.). Each tile links to the filtered page.
- Quick actions row: Add stock, Upload Excel (-> /pharmacy/add-stock?tab=excel), New counter sale, Open queue.
- Sections: live Pending Queue (top 5, from usePharmacyQueueLive), Low stock (top 5 with "Reorder" -> Add Stock pre-filled with that medicine), Expiring soon (top 5 with days left), Recent stock movements (last 10), Top dispensed medicines this week (small bar chart via existing Recharts setup and chartTheme), Today's bills.
- Loading skeletons, friendly empty states, and an error state with Retry for each section independently (one failing query must not blank the page).
- Greeting with pharmacist name and clinic name; date in IST.

Counter Billing check

- Verify /pharmacy/billing works end-to-end for a pharmacist: pick medicines (only in-stock, non-expired, FEFO batch), quantities, discount, payment mode, save. Saving must reduce stock (through an RPC or dispense path, not a client loop) and the bill must appear in Today's bills. If the current implementation does not reduce stock, add `create_pharmacy_counter_sale(input jsonb)` RPC that creates the invoice, invoice items, payment record and stock movements atomically.

Task 9 - Demo data that makes a great demo

Extend `supabase/seed/demo_seed.sql` (keep it idempotent, keyed on the demo clinic slug, dated relative to current_date, and keep the existing fixed medicine ids 33333333-...-000000000001 to ...010 because prescriptions reference them). Section 2 already copies role permissions into staff_permissions - after Task 1 this includes pharmacy.manage; confirm it.

Data	What to seed
Medicines	40 total (keep the 10, add 30) with realistic Indian brands and generics across categories: antibiotics (Azithromycin 500, Cefixime 200, Augmentin 625), analgesics (Dolo 650, Combiflam, Diclofenac gel), GI (Pan-D, Ondansetron 4, Rantac), allergy (Montelukast-LC, Levocetirizine), diabetes/BP/cardiac (Glimepiride 1, Telmisartan 40, Atorvastatin 10, Ecosprin 75), syrups (Ascoril, Calpol), dental (Chlorhexidine, Sensodyne Repair), derma (Clotrimazole, Mupirocin, Sunscreen SPF50), supplements (Shelcal 500, Becosules, Neurobion Forte), ORS, Insulin pen. Fill generic, strength, unit, manufacturer, HSN, GST, reorder level.
Batches	1-3 per medicine (~70) with a deliberate mix: ~70% healthy, 6 low stock, 3 out of stock (qty 0 / depleted), 5 expiring within 30 days, 3 within 31-90 days, 3 already expired. Realistic purchase/selling/MRP.
Purchases	8 stock_purchases over the last 60 days from 3 suppliers (MedSupply Distributors, Karnataka Pharma Agencies, Apollo Wholesale), metadata.source manual or excel, each linked to its batches, with matching 'add_stock' stock_movements.
Prescriptions -> orders	Today: 6 routed prescriptions from the 3 doctors -> 6 pending orders (one has an item with no catalogue match = unavailable, one has an out-of-stock item, one partially dispensed). Last 14 days: ~20 dispensed orders with 'dispense' stock movements and dispensed_by = pharmacist. Prescription item names must match catalogue names exactly so matching works.
Billing	Pharmacy invoices for the dispensed orders plus ~25 counter sales spread across 14 days (cash/UPI/card mix) so today's sales and the weekly chart are non-empty. Today: at least 5 bills.
Movements	A few 'damage', 'expired' and 'correction' adjustments with reasons, so the history drawer looks real.
Alerts	Recompute low_stock_alerts at the end, as the seed already does.

- Update the final summary query at the bottom of the seed to also print counts of medicines, batches, stock_purchases, pharmacy_orders by status and pharmacy invoices.
- The seed must still run cleanly on a database where only some demo Auth users exist (skip what depends on missing users, as it does today).
- Update the README Demo section: how to create the Auth users, apply migrations, run the seed, then the 5-minute demo script below.

Task 10 - Documentation and final verification

- Write `docs/PHASE_16_PHARMACIST_MODULE.md`: root causes found, migrations added, RPCs (inputs/outputs/permissions), new pages, Excel template spec, demo steps.
- Run `npm run build` and `npm run lint` in frontend/ - both must pass with zero errors.
- List for the user, in order, every SQL file to apply and the seed to re-run.

5-minute demo script (must work after the seed)

1. Sign in as VNX-PHAR-001 -> dashboard shows 6 pending prescriptions, today's sales, low-stock and expiry tiles with real numbers.
2. Open Prescription Queue -> open a pending order -> see allergies, FEFO batches -> Dispense and create bill.
3. In another browser, sign in as doctor@vernex.test, complete a consultation with medicines -> the pharmacist screen shows the new prescription live.
4. Add Stock -> Excel tab -> Download template, then upload docs/sample-data/pharmacy_stock_upload_sample.csv -> preview -> import -> stock and low-stock tiles update.
5. Add Stock -> Manual tab -> receive 100 more of an existing batch -> Medicine Stock shows the summed quantity and movement history.
6. Counter Billing -> quick sale -> appears in Today's bills.

Definition of done

Pharmacist login works by email and staff ID; no page the pharmacist can see in the sidebar ever bounces them to login or Access Denied; every doctor prescription sent to pharmacy appears for the pharmacist automatically and live; stock can be added manually and by Excel/CSV with preview and validation, atomically, with quantities added (never overwritten); demo data makes every pharmacist screen look full and realistic; build and lint pass; docs and README updated. Report back with what was verified for each task and anything that could not be verified.